

EK Model

→ setup for hat algebra; derivation begins on page 3

1) Parameters, Variables, Model Design

key equations highlighted in green

a) Parameters

T_i : state of tech / absolute advantage in country i ($\uparrow T_i$ shifts up efficiency distribution)

↳ i.e., higher T_i indicates higher likelihood for country i to be productive at any good

Θ : dispersion in productivity / comparative advantage

↳ lower $\Theta \rightarrow$ greater dispersion in productivity draws across goods (vice versa for $\uparrow \Theta$)

d_{ni} : geographic trade barriers in shipping goods from country i to n

↳ let $d_{ii} = 1$; $d_{ni} > 1$ for $n \neq i$

b) Variables and Notation

N : number of countries

L_i : labour force in country i

D_n : international transfers; $\sum_n D_n = 0$; if $D_n > 0$, trade deficit; if < 0 , trade surplus

$h(w)$: numeraire fn for transfers; e.g., $h(w) = w US$

w_i : wage in country i

subscript n_i : country n imports (buyer), country i exports (seller)

X_n : total spending of country n

X_{ni} : total spending of country n on goods from i

$\pi_{ni} = \frac{X_{ni}}{X_n}$ share of n 's spending on goods from i

c) Model Design

1) $\exists$ a continuum of goods, represented by $[0, 1]$

2) production fn takes in only labour, so $Y_i = w_i L_i$

3) each country randomly draws good-specific productivity, $Z_i(j)$ from a Fréchet distribution

↳ country i 's efficiency in producing good j is given by $Z_i(j)$ with $Z_i \sim \text{Fréchet}(T_i, \Theta)$

↳ $F_i(z) = P(Z_i \leq z) = e^{-T_i z^{-\Theta}}$ the cdf of country i 's productivity, $T_i > 0$, $\Theta > 0$

4) this version of the model has a single sector only (i.e., manufacturing), so the cost of production, given 2), is simply: $c_i = w_i$

↳ the productivity-adjusted cost of producing good j in country i is then: $\frac{w_i}{Z_i(j)}$

5) iceberg costs imply $P_{ni}(j) = (w_i d_{ni}) / Z_i(j)$

↳ using 3), the RV representing the price of buying good j from country i is given by: $P_{ni} = (w_i d_{ni}) / Z_i$

6) by assumption of perfect competition, buyers purchase from the seller with lowest price: $P_n = \min \{ P_{ni}; i, \dots, N \}$

7) given the distribution of $Z_i \sim \text{Fréchet}$, can derive distribution of P_{ni} to represent the randomness in price that country i charges to deliver a good to country n
 $\hookrightarrow \Phi_n = \sum_{i=1}^N T_i (c_i d_{ni})^{-\theta}$ gives how favourable the overall price environment is (high $\Phi \rightarrow$ low prices)

8) it is with probability $\pi_{ni} = \frac{T_i (c_i d_{ni})^{-\theta}}{\Phi_n}$ that country i has the lowest offer
 $\hookrightarrow$ gives the fraction of goods Φ_n country n purchases from i

9) consumers purchase amount $Q(j)$ to maximize CES objective:

$$U = \left[\int_0^1 Q(j)^{(\sigma-1)/\sigma} dj \right]^{\sigma/(\sigma-1)} \quad \text{where } \sigma \text{ is the elasticity of substitution}$$

s.t. country n 's total spending, X_n

10) putting 7) and 8) together and using 4) (i.e., $c_i = w_i$) get:

$$\pi_{ni} = \frac{X_{ni}}{X_n} = \frac{T_i (w_i d_{ni})^{-\theta}}{\Phi_n} = \frac{T_i (w_i d_{ni})^{-\theta}}{\sum_{k=1}^N T_k (w_k d_{nk})^{-\theta}} = \frac{\text{country } n\text{'s total spend. on goods from } i}{\text{country } n\text{'s total spend}}$$

11) given 2) and adding trade imbalances as per assignment, get $X_n = w_n L_n + h/w_n D_n$

12) market clearing condition implies that for each country, total wage bill = total revenue
 $\hookrightarrow$ by perf. competition assumption profits = 0 and have CES with labour-only production
 $\hookrightarrow$ so, every dollar spent on i 's goods by any importer n must equal i 's labour income
 $\hookrightarrow w_i L_i = \sum_{n=1}^N \pi_{ni} X_n$

13) since $\pi_{ni} \cdot X_n = X_{ni} \rightarrow w_i L_i = \sum_{n=1}^N X_{ni}$

14) $y_n^o = \frac{\sum_i X_{in}^o}{\sum_i X_{ni}^o} = \frac{\text{total sales of country } n}{\text{total spending of country } n} = \frac{\text{income}}{\text{spending}} = \frac{w_n^o L_n^o}{X_n^o} \rightarrow y_n^o \cdot X_n^o = w_n^o L_n^o$

2. Hat Algebra Derivation

a) main building blocks in level form

1) $X_n = w_n L_n + h(w) D_n \rightarrow$ total spending with trade imbalances

2) $\pi_{ni} = \frac{X_{ni}}{X_n} = \frac{T_i(w_i d_{ni})^{-\theta}}{\Phi_n} \rightarrow$ trade shares; given wage vector, what fraction of country n 's spending goes to each i

$$\Phi_n = \sum_{k=1}^N T_k(w_k d_{nk})^{-\theta}$$

3) $w_i L_i = \sum_{n=1}^N \pi_{ni} X_n \rightarrow$ market clearing condition

sub-in equations 1 and 2 into 3 to get levels system:

$$w_i L_i = \sum_{n=1}^N \left(\frac{T_i(w_i d_{ni})^{-\theta}}{\sum_{k=1}^N T_k(w_k d_{nk})^{-\theta}} \right) (w_n L_n + h(w) D_n), \quad i = 1, 2, \dots, N \quad \text{eq. A}$$

Thus, in levels, equilibrium wages solve a nonlinear system of N (one such equation per country) equations with N unknowns (RHS is a fn of the wage vector $(w_1, \dots, w_N)$)

Rather than solving this system in levels, which would require the underlying levels of T_i and d_{ni} , exact hat algebra expresses the counterfactual equilibrium in changes relative to the observed baseline.

Next, express all conditions in changes relative to the observed baseline.

Let ' denote new/counterfactual values and 0 denote baseline values

hat variable = ratio of new to old $\rightarrow \hat{x} = \frac{x'}{x^0}$

b) Simplify the clearing condition

as described in 13), have that share $\cdot$ spending is equivalent to dollar flow, so

$$w_i L_i = \sum_{n=1}^N \pi_{ni} X_n = w_i L_i = \sum_{n=1}^N X_{ni} \quad \text{since} \quad \pi_{ni} = \frac{X_{ni}}{X_n} \quad \text{and thus} \quad X_{ni} = \pi_{ni} X_n$$

then, by hat algebra notation, define:

$$w_i' L_i' = \sum_{n=1}^N X_{ni}' \quad \text{and} \quad w_i^0 L_i^0 = \sum_{n=1}^N X_{ni}^0$$

so, $\hat{w}_i \hat{L}_i = \frac{w_i' L_i'}{w_i^0 L_i^0}$

and $w_i' L_i' = w_i^0 L_i^0 \cdot \hat{w}_i \hat{L}_i = \sum_{n=1}^N X_{ni}'$

so, $\hat{w}_i \hat{L}_i = \frac{\sum_{n=1}^N X_{ni}'}{w_i^0 L_i^0}$

$$\hat{w}_i \hat{L}_i = \sum_{n=1}^N \frac{X_{ni}^0 \cdot \hat{X}_{ni}}{w_i^0 L_i^0}$$

$$\hat{w}_i \hat{L}_i = \sum_{n=1}^N \frac{X_{ni}^0}{\sum_{d=1}^N X_{di}^0} \cdot \hat{X}_{ni} \quad \text{eq. B}$$

want to write $\hat{w}_i \hat{L}_i$ w/o use of ' values as these are unobserved / currently unknown. use $X_{ni}' = X_{ni}^0 \cdot \hat{X}_{ni}$ can bring $w_i^0 L_i^0$ into sum as constant

because $w_i^0 L_i^0 = \sum_{d=1}^N X_{di}^0$ by clearing condition labour income = revenue

c) derive $\hat{X}_{ni}$

now having written the LHS of eq. A in nat algebra form (eq. B) continue to write terms out. since $X_{ni}' = \pi_{ni} X_n$ it follows that $\hat{X}_{ni} = \frac{\hat{\pi}_{ni}}{\hat{X}_n}$ thus it remains to derive $\hat{\pi}_{ni}$ and $\hat{X}_n$

c.i) derive $\hat{\pi}_{ni}$

define:

$$\pi_{ni}^0 = \frac{T_i^0 (w_i^0 d_{ni}^0)^{-\theta}}{\phi_n^0} \quad \pi_{ni}' = \frac{T_i' (w_i' d_{ni}')^{-\theta}}{\phi_n'}$$

$$\begin{aligned} \text{then, } \hat{\pi}_{ni} &= \frac{\pi_{ni}'}{\pi_{ni}^0} = \frac{T_i' (w_i' d_{ni}')^{-\theta}}{\phi_n'} \cdot \frac{\phi_n^0}{T_i^0 (w_i^0 d_{ni}^0)^{-\theta}} \\ &= \frac{T_i'}{T_i^0} \cdot \frac{(w_i' d_{ni}')^{-\theta}}{(w_i^0 d_{ni}^0)^{-\theta}} \cdot \frac{\phi_n^0}{\phi_n'} \end{aligned}$$

$$\therefore \hat{\pi}_{ni} = \hat{T}_i (\hat{w}_i \hat{d}_{ni})^{-\theta} \cdot \frac{1}{\hat{\phi}_n} \quad \text{eq. C}$$

c.ii) derive $\hat{\Phi}_n$

$$\Phi_n = \sum_{k=1}^N T_k (w_k d_{nk})^{-\theta}$$

let $\alpha_k = T_k (w_k d_{nk})^{-\theta}$ be each term in the sum so:

$$\Phi_n = \sum_{k=1}^N \alpha_k \quad \text{and} \quad \alpha'_k = \hat{\alpha}_k \cdot \alpha_k^\circ \quad \text{and} \quad \hat{\alpha}_k = \hat{T}_k (\hat{w}_k \hat{d}_{nk})^{-\theta} \quad \text{then,}$$

$$\begin{aligned} \Phi'_n &= \sum_{k=1}^N \alpha'_k = T'_k (w'_k d_{nk}')^{-\theta} \\ &= \sum_{k=1}^N \alpha_k^\circ \cdot \hat{T}_k (\hat{w}_k \hat{d}_{nk})^{-\theta} \\ &= \sum_{k=1}^N \left[T_k^\circ (w_k^\circ d_{nk}^\circ)^{-\theta} \right] \cdot \hat{T}_k (\hat{w}_k \hat{d}_{nk})^{-\theta} \end{aligned}$$

then, use Φ'_n expression $\div \Phi_n^\circ$ to get $\hat{\Phi}_n$.

$$\hat{\Phi}_n = \sum_{k=1}^N \frac{T_k^\circ (w_k^\circ d_{nk}^\circ)^{-\theta}}{\Phi_n^\circ} \cdot \hat{T}_k (\hat{w}_k \hat{d}_{nk})^{-\theta} = \pi_{nk}^\circ$$

then,

$$\hat{\Phi}_n = \sum_{k=1}^N \pi_{nk}^\circ \cdot \hat{T}_k (\hat{w}_k \hat{d}_{nk})^{-\theta} \quad \text{eq. D}$$

$\therefore$ plugging eq. D back into eq. C gives:

$$\hat{\pi}_{ni} = \frac{\hat{T}_i (\hat{w}_i \hat{d}_{ni})^{-\theta}}{\sum_{k=1}^N \pi_{nk}^\circ \cdot \hat{T}_k (\hat{w}_k \hat{d}_{nk})^{-\theta}} \quad \text{eq. E}$$

c.iii) derive $\hat{X}_n$

$X_n = w_n L_n + h(w) D_n$ so define:

$$X_n^\circ = w_n^\circ L_n^\circ + h(w^\circ) D_n^\circ \quad \text{and} \quad X'_n = w'_n L'_n + h(w') D'_n$$

using: $w'_n = w_n^\circ \cdot \hat{w}_n$, $L'_n = L_n^\circ \cdot \hat{L}_n$, $D'_n = D_n^\circ \cdot \hat{D}_n$ gives:

$$X'_n = w_n^\circ \hat{w}_n L_n^\circ \hat{L}_n + h(w') D_n^\circ \hat{D}_n$$

$$\text{then assemble } \hat{X}_n = \frac{X'_n}{X_n^\circ} = \frac{w_n^\circ L_n^\circ}{X_n^\circ} \hat{w}_n \hat{L}_n + \frac{h(w') D_n^\circ}{X_n^\circ} \hat{D}_n$$

$$\text{by 14), have that} \quad y_n^\circ = \frac{\sum_i X_{ni}^\circ}{\sum_i X_{ni}^\circ} = \frac{w_n^\circ L_n^\circ}{X_n^\circ}$$

$$\text{so, } \hat{X}_n = y_n^0 \hat{w}_n \hat{L}_n + \frac{h(w') D_n^0}{X_n^0} \hat{D}_n$$

and note $h(w') D_n^0 = X_n^0 - w_n^0 L_n^0$ by definition of X_n^0
 $= (1 - y_n^0) X_n^0$

$$\text{so } \frac{h(w') D_n^0}{X_n^0} = \frac{h(w')}{h(w^0)} \cdot \frac{h(w^0) D_n^0}{X_n^0} = \frac{h(w')}{h(w^0)} (1 - y_n^0)$$

$$\text{then, } \hat{X}_n = y_n^0 \hat{w}_n \hat{L}_n + \frac{h(w')}{h(w^0)} (1 - y_n^0) \hat{D}_n$$

and using $w' = w^0 \hat{w}$ and $h(w') = h(w^0 \hat{w})$:

$$\therefore \hat{X}_n = y_n^0 \hat{w}_n \hat{L}_n + \frac{h(w^0 \hat{w})}{h(w^0)} (1 - y_n^0) \hat{D}_n \quad \text{eq. F}$$

d) Combine

recall $\hat{X}_{ni} = \hat{\pi}_{ni} \cdot \hat{X}_n$. plugging in eq. E for $\hat{\pi}_{ni}$ and eq. F for $\hat{X}_n$ gives:

$$\hat{X}_{ni} = \frac{\hat{T}_i(\hat{w}; \hat{d}_{ni})^{-\theta}}{\sum_{k=1}^N \hat{\pi}_{nk}^0 \cdot \hat{T}_k(\hat{w}; \hat{d}_{nk})^{-\theta}} \left(y_n^0 \hat{w}_n \hat{L}_n + \frac{h(w^0 \hat{w})}{h(w^0)} (1 - y_n^0) \hat{D}_n \right)$$

then plugging in the above expression for $\hat{X}_{ni}$ into eq. B gives:

$$\hat{w}_i \hat{L}_i = \sum_{n=1}^N \frac{X_{ni}^0}{\sum_{d=1}^N X_{di}^0} \cdot \frac{\hat{T}_i(\hat{w}; \hat{d}_{ni})^{-\theta}}{\sum_{k=1}^N \hat{\pi}_{nk}^0 \cdot \hat{T}_k(\hat{w}; \hat{d}_{nk})^{-\theta}} \left(y_n^0 \hat{w}_n \hat{L}_n + \frac{h(w^0 \hat{w})}{h(w^0)} (1 - y_n^0) \hat{D}_n \right)$$

rearranging this to excess demand form:

$$Z_i(\hat{w}) = \sum_{n=1}^N \frac{X_{ni}^0}{\sum_{d=1}^N X_{di}^0} \cdot \frac{\hat{T}_i(\hat{w}; \hat{d}_{ni})^{-\theta}}{\sum_{k=1}^N \hat{\pi}_{nk}^0 \cdot \hat{T}_k(\hat{w}; \hat{d}_{nk})^{-\theta}} \left(y_n^0 \hat{w}_n \hat{L}_n + \frac{h(w^0 \hat{w})}{h(w^0)} (1 - y_n^0) \hat{D}_n \right) - \hat{w}_i \hat{L}_i$$

$$Z_i(\hat{w}) = 0, \quad i = 1, 2, \dots, N$$

given shocks to trade costs $\hat{d}_{ni}$, tech $\hat{T}_i$, and transfers $\hat{D}_n$, the counterfactual wage changes $\hat{w}$ are determined by the N nonlinear market clearing conditions $Z_i(\hat{w}) = 0$